

CRRT-UDE : context-gated regulatory transport as a Universal Differential Equation

Receiver state $z \in \mathbb{R}^{773}$

14 PROGENy pathways
+ ~759 CollecTRI TF activities

Spatial context w

DestVI composition, BANKSY
niche summaries of the neighborhood

Regulatory sub-vector $r(z)$

TF/pathway subset selected by r_{idx}

Field decomposition
self dynamics (structured + residual)

$$v_{\text{self}} = A z + b + B r(z) + \rho_s(z, \tau)$$

$$g(w) = \tanh(G w + c)$$

learned gate: how the niche modulates each regulator

$$v_{\text{context}} = C [r(z) \odot g(w)] + \rho_c(z, w, \tau)$$

context dynamics (niche $\times$ regulator interaction)

Universal DE

$$\frac{dz}{d\tau} = v_{\text{self}} + v_{\text{context}}$$

neural ODE integrated $\tau : 0 \rightarrow 1$

$d_\rho \approx 0.5$: structure-dominant
(mechanistic terms, not a black box)

Trajectories & endpoints

integrated $z(\tau)$; conditional Sinkhorn validation

R_{cond} per-cell

context-conditioned transport contrast
= how the niche redirects the transition

Helmholtz-Hodge flux

$v = -\nabla \Phi + v_{\text{rot}}$
rotational fraction = non-equilibrium index

Three-stage OT-CFM training (conditional-stratified coupling; patient-held-out 5-fold CV)

Stage A

fit v_{self}
OT-CFM, context off

Stage B

freeze v_{self}
fit v_{context}

Stage C

anchored
joint refinement